

## Auxiliar calculus based on Beal's conjecture

Beal's equation

$$A^x + B^y = C^z$$

i converted the problem statement into

$$(p * c_1)^x + (p * c_2)^y = (p * c_3)^z$$

A is  $(p * c_1)$  B is  $(p * c_2)$  and C is  $(p * c_3)$

Throughout, p denotes any prime number.  $c_1, c_2$  and  $c_3$ , represent composite numbers, bigger than 1. P also bigger than 1.

when i do  $A^x$ , means  $p^x * c_1^x$  (or  $(p * c_1)^x$ ), which is equivalent

divide equation by C

$$\frac{A^x}{C} + \frac{B^y}{C} = C^{z-1}$$

convert A,B and C, to my form

$$\frac{p^x * c_1^x}{p * c_3} + \frac{p^y * c_2^y}{p * c_3} = C^{z-1}$$

$\frac{p^x}{p}$  equal  $p^{x-1}$  and  $\frac{p^y}{p}$  denotates  $p^{y-1}$

$$p^{x-1} * \frac{c_1^x}{c_3} + p^{y-1} * \frac{c_2^y}{c_3} = C^{z-1}$$

multiply both sides by  $c_3$

$$p^{x-1} * c_1^x + p^{y-1} * c_2^y = C^{z-1} * c_3$$

divide both sides by  $C^{z-1}$

$$\frac{p^{x-1} * c_1^x}{C^{z-1}} + \frac{p^{y-1} * c_2^y}{C^{z-1}} = c_3$$

multiply both sides, by  $C^{z-1}$

$$p^{x-1} * c_1^x + p^{y-1} * c_2^y = c_3 * C^{z-1}$$

divide both sides by  $p^{x-1} * c_1^x + p^{y-1} * c_2^y$

$$1 = \frac{c_3 * C^{z-1}}{p^{x-1} * c_1^x + p^{y-1} * c_2^y}$$

multiply C on both sides

$$C = \frac{C^z * c_3}{(p^{x-1} * c_1^x + p^{y-1} * c_2^y)}$$

divide both sides by  $c_3$

$$\frac{C}{c_3} = \frac{C^z}{(p^{x-1} * c_1^x + p^{y-1} * c_2^y)}$$

develop left side

$$\frac{p * c_3}{c_3} = \frac{C^z}{(p^{x-1} * c_1^x + p^{y-1} * c_2^y)}$$

cancel  $c_3$  on left side.

$$p = \frac{C^z}{(p^{x-1} * c_1^x + p^{y-1} * c_2^y)}$$